

Housing Policy, Portfolio Risk, and the Home-Purchase Threshold: Financial Risk-Taking among Young Prospective Homebuyers

Presenter: Mihyun You

Department of Economics
Georgetown University

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Introduction

Motivation

Young Americans are taking large financial risks.

Investment attitude / behavior	Age group		
	18–34	35–54	55+
Need to take big risks to reach financial goals	62%	42%	16%
Traded options	43%	24%	10%
Purchased meme / viral investments	29%	18%	2%
Invested in cryptocurrency ^a	50%	36%	10%

^a Among respondents aware of cryptocurrency.

Source: FINRA Investor Education Foundation, 2024 Investor Survey.

Survey respondents have investments outside of retirement accounts.

► Are they simply risk-lovers?

What if housing market is part of the reason?

The home-purchase wealth hurdle has risen across generations.

Home-purchase hurdle	Age 25–34		
	1985	2000	2025
Median weekly earnings	\$349	\$549	\$1,143
Annualized earnings ^a	\$18.1K	\$28.5K	\$59.4K
Median new-home price	\$84.3K	\$169.0K	\$417.4K
Home price / annualized earnings	4.65×	5.92×	7.02×
20% benchmark down payment	\$16.9K	\$33.8K	\$83.5K
20% down payment / annualized earnings	93%	118%	140%

^a Median usual weekly earnings multiplied by 52.

Notes: Earnings are for full-time wage and salary workers, ages 25–34, both sexes. Home price is the median sales price of new houses sold. A 20% down payment is an illustrative benchmark, not a universal mortgage requirement.

Sources: U.S. Bureau of Labor Statistics, Current Population Survey; U.S. Census Bureau/HUD, New Residential Sales.

Related Literature

Recent research connects housing attainability to risk-taking behavior.

1. **Housing attainability $\Rightarrow$ Portfolio risk-taking**

(Lee and Yoo, 2025; Dai, Savor, and Wilson, 2025; Arkolakis et al., 2026)

2. **Financial wealth/housing policy $\Rightarrow$ Housing-market equilibrium**

(Aiello et al., 2023; Abramson and Landvoigt, 2025)

This project:

Housing policy $\Rightarrow$ Portfolio response $\Rightarrow$ Housing demand $\Rightarrow$
House prices

Research Question

- Does endogenous portfolio reoptimization change the equilibrium transmission of housing policy?

Conventional housing policy channel:

$$\text{Housing policy} \rightarrow D^H, S^H \rightarrow P$$

This paper:

$$\text{Housing policy} \rightarrow x^* \rightarrow \Pr(a' \geq dP) \rightarrow D^H \rightarrow P^H$$

where x^* : endogenous optimal share invested in the risky asset, a' : next-period liquid financial wealth after portfolio returns, d : required down-payment share.

Quantitative Model

Two-Period Prototype Model

Date 0

Date 0: portfolio choice

$a \sim F(a)$, $P, R \sim F(R)$ known.

$$x^*(a; P) = \arg \max_{x \in [0,1]} E_R \underbrace{[V_1(a'(R), P)]}_{\text{best attainable value of next-period}}$$

where $V_1(a', P)$: continuation value from the optimal housing decision given realized liquid wealth a' .

$$V_1(a', P) = \begin{cases} V_1^R(a', P), & \text{if } a' < dP, \\ \max \{ V_1^R(a', P), V_1^H(a', P) \}, & \text{if } a' \geq dP. \end{cases}$$

where $V_1^R(a', P)$: value of remaining a renter, $V_1^H(a', P)$: value of purchasing the home.

Two-Period Prototype Model

Date 1

Date 1: home-purchase feasibility

Once R realizes, realized liquid wealth is

$$a' = a[(1 - x)R_f + xR]$$

where a : initial liquid wealth, $x \in [0, 1]$: speculative portfolio share, R_f : gross rate of return of safe asset, R : gross rate of return of risky asset with

$$E[R] < R_f.$$

Home purchase is feasible if

$$a' \geq d \times P$$

where d : required down-payment share, P : equilibrium house price.

Portfolio Choice and Housing Demand

Portfolio risk changes the probability of crossing the down-payment threshold.

$$q(a, x; P) = \Pr[a((1 - x)R_f + xR) \geq d \times P].$$

Then,

$$\Pr(\text{buy} \mid a) = \underbrace{q(a, x^*(a; P); P)}_{\text{Can buy}} \underbrace{\pi_H(P)}_{\text{Choose to buy}}$$

where $\pi_H(P)$: purchase probability conditional on feasibility.

Aggregate demand:

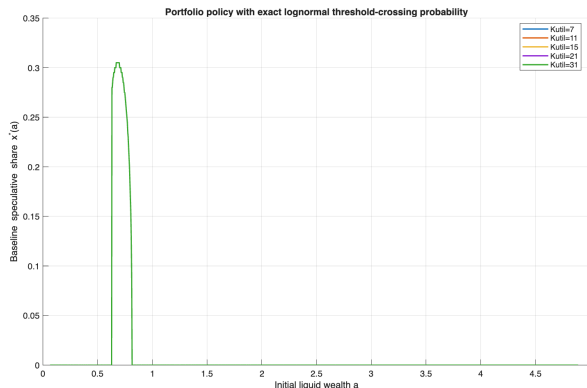
$$D^H(P) = \int q(a, x^*(a; P); P) \pi_H(P) dF(a).$$

In equilibrium,

$$D^H(P) = S^H(P).$$

Home-Purchase Threshold Mechanism

Risk-taking is concentrated near the home-purchase threshold.



Too far
 $x^*(a) = 0$

Within reach
 $x^*(a) > 0$

Safely attainable
 $x^*(a) = 0$

Identifying the Portfolio-Feedback Effect

Partial Equilibrium (No Portfolio Feedback):

$$x = x_0^*(a)$$

General Equilibrium (Portfolio Feedback):

$$x = x^*(a; P, \text{policy})$$

We can calculate

$$\Delta P_{\text{portfolio}}^H = \Delta P_{GE}^H - \Delta P_{PE}^H, \quad \mathcal{A} = \frac{\Delta P_{GE}^H}{\Delta P_{PE}^H}.$$

where ΔP represents the house price change after the relevant policy.

Follow-Up Plan

What Next?

- ▶ Study literature
- ▶ Policy comparison:
down-payment assistance vs. housing supply
- ▶ If portfolio-feedback effect found meaningful, parsimonious
lifecycle model (endogenous a)

References I

-  Aiello, Darren et al. (2023). *The effects of cryptocurrency wealth on household consumption and investment*. Tech. rep. National Bureau of Economic Research.
-  Abramson, Boaz and Tim Landoigt (2025). “Curbing rising housing costs: A model-based policy comparison”. In: *Journal of Economic Perspectives* 39(3), pp. 27–44.
-  Dai, Bingcun, Pavel G Savor, and Mungo Ivor Wilson (2025). “All or Nothing: Indivisible Goods, Relative Wealth Concerns, and Attitudes to Risk”. In.
-  Lee, Seung Hyeong and Younggeun Yoo (2025). “giving up”: The impact of decreasing housing affordability on consumption, work effort, and investment”. In.
-  Arkolakis, Costas et al. (2026). “Gambling for Goals”. In.